EXPLORING BUSINESS · CHAPTER 7

From Product Design and Development...

Products and types of new products, ideas and opportunities, utilities, industry, market, segment and niche, demand forecasting, breakeven analysis (not for the MCQ), the 7-step development process, AI and patents.

01 The common thread: the PowerSki Jetboard

The chapter follows **Bob Montgomery**, who designed the **PowerSki Jetboard**: a motorized surfboard powered by a 45-horsepower engine that weighs only 45 pounds, so you can "surf" without waves. It is the textbook case of going *from product design and development...*: an idea, a study of the industry and of demand, a profitability check, development in several steps, and protection through patents.

HOW THE CHAPTER IS BUILT

1. What is a **product**, and what kinds of new products are there?
2. Where do **ideas** come from, and when do they become **opportunities**?
3. Understanding your **industry** (industry → market → segment → niche).
4. Forecasting **demand** and **market share**.
5. Calculating the **breakeven point** (not for the MCQ).
6. **Developing** the product in 7 steps, with the help of AI.
7. **Protecting** the idea with a patent.

02 What is a product?

Product: something that can be **marketed** to customers because it provides them with a **benefit** and satisfies a **need**.

A product can be a **good** (tangible), a **service** (intangible) or a combination of both: when you buy an Apple computer, you get a good (the computer) and services (warranty, support). An in-house executive training program is a service.

The four types of product development

TYPE	DEFINITION	EXAMPLE (JUST BORN, POWERSKI)
New-to-the-company product	New to the company, but already sold by a competitor in the past	Peeps marshmallow Easter decorating kits
Improvement in an existing product	An enhancement of a product already on the market	Mike & Ike's: new ingredients and resealable pouches → sales up 20%
Extension of an existing product line	A new product developed as a variation of an existing one	Peeps inside a chocolate egg
New-to-the-market product	Has never been available to consumers or manufacturers	The PowerSki Jetboard

NEW TO THE MARKET: RISKY BUT RARE

These products make up only about **10%** of new products. They are the riskiest (no proven market), but the company enjoys a **temporary monopolistic position** until competitors catch up.

The entrepreneurial start-up

The Jetboard came from an **entrepreneurial start-up**, which has **four characteristics**:

- **innovative** products and/or practices;
- goals of **profitability and growth**;
- a focus on **new opportunities**;
- owners who are **willing to take risks**.

CALCULATED, NOT UNNECESSARY, RISK

Entrepreneurship is about **carefully calculated risks**, not unnecessary risks. Decisions improve with input from two sources: **information gathered from research** and **knowledge gained from personal experience**.

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03 Where do product ideas come from?

The right question is not "*What do I want to sell?*" but "**What does the customer want to buy?**"

> *"The best way to have a good idea is to have lots of ideas."* — ***Linus Pauling***

About **30,000** new food, beverage and beauty products are launched each year; up to **90%** fail within a year. Hence the importance of a truly remarkable idea.

Purple cow (Seth Godin): a **truly remarkable product** that stands out like a purple cow in a herd of ordinary cows. Examples: **Dean Kamen** (Segway, wearable infusion pump); **Jacob Dunnack**, who invented the **JD Batball** (a bat that stores the balls) at age six.

Where do companies look?

INSIDE THE ORGANIZATION

Marketing, sales, research, manufacturing; some firms create **internal entrepreneurial units**.

OUTSIDE THE ORGANIZATION

Customers and the **competition**; creativity consultants such as **Doug Hall** (Eureka! Ranch).

Most product ideas come from **entrepreneurs and small-business owners**, but medium and large organizations must also find product-development opportunities to remain competitive.

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04 From idea to opportunity: the four utilities

Business opportunity: an idea turns into an opportunity when it has **commercial potential**, that is, when you can make money by selling the product.

Not every idea is an opportunity: Bic underwear, Harley-Davidson perfume, bottled water for pets and Colgate kitchen entrees all flopped.

Creating value: the utilities

UTILITY	THE PRODUCT PROVIDES VALUE BY...	EXAMPLE
Time utility	being available at a **convenient time**	Bottled water sold at an outdoor concert
Place utility	being available in a **convenient location**	A hot-dog vendor outside an office building
Ownership utility	**transferring ownership** of the product	A real estate agent helping a family buy a house
Form utility	**changing the composition** of the product	Apparel made from fabric; liquid rather than powdered detergent

THREE QUESTIONS BEFORE GOING AHEAD (FIGURE 7.6)

- **Who** would my customers be?
- **Why** would customers buy the product from me?
- **How** will customers benefit from my product?

05 Understand your industry: industry, market, segment, niche

Before developing a product, you need to understand the industry in which it will be sold. You start broad and **progressively narrow** your focus.

LEVEL	DEFINITION	EXAMPLE: FOOTWEAR
Industry	A group of related businesses that do similar things and **compete** with each other	The footwear industry
Market	A group of buyers who share a **need**	Athletic (jogging) shoes
Market segment	A smaller group of customers with **common characteristics** that influence their buying decisions	Joggers over 65 living in the Southeast
Niche	A narrowly defined group with **specific needs** (an unmet need)	Active adults in Florida retirement communities

Other examples from the book: **computer** industry → laptop buyers → college students wanting a hybrid laptop/tablet → students who can afford a high-end hybrid with gaming capabilities; **recreational activities** → leisure travelers → people who take cruises → college students who want cruises.

The three segmentation variables

DEMOGRAPHIC

Who the customer is: age, sex, income, education, ethnicity, occupation, family life stage (young marrieds, full nesters, empty nesters).

GEOGRAPHIC

Where the customer lives: region, climate, city size, urban / suburban / rural, world region (North America, Western Europe, Pacific Rim).

PSYCHOGRAPHIC

How the customer lives and thinks: lifestyle, activities, interests, personality, attitude toward risk (achievers / strivers / strugglers, compulsive, gregarious, ambitious, economy-minded).

COMMON TRAPS

Occupation (blue-collar, white-collar) and **age** are **demographic**. **Social class** is usually grouped with lifestyle, so **psychographic** (Kotler's classification); some textbooks treat it as demographic. Attitude toward **risk** is **psychographic**. **Climate** is **geographic**. In the book's example, "older people" is demographic, "living in the Southeast" geographic and "who jog" psychographic.

Assessing your competition

Questions to ask: is the industry **growing or contracting**? Who are your major **competitors**? What **opportunities and threats** exist? Has the industry changed recently? How important is **technology**? Is the industry **mature**, or are new companies entering it? Do companies make **reasonable profits**? Sources: competitors themselves, the internet, people in the industry, customers.

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06 Forecasting demand and market share

Before investing, the key question is: *are there enough customers willing to buy my product at a price that will allow me to make a profit?* Forecasting demand is one of the hardest tasks in business.

IDENTIFYING YOUR CUSTOMERS (FIGURE 7.8)

- Who are my **primary customers**?
- Will I sell to **individuals**, **businesses**, or both?
- If I sell to businesses, who are the **end users**?

Three approaches, alone or combined

APPROACH	EXAMPLE FROM THE BOOK
Talk with **people in similar businesses**	Opening a cartoon-themed pizza parlor: talk to pizza parlor owners elsewhere
Talk with **potential customers**	Opening a frozen yogurt shop in North Dakota: ask residents if they would buy it in winter
Examine **published industry data**	National Sporting Goods Association reports to estimate the total market

Market share: the company's **portion of the targeted market**.

EXAMPLE: JOGGING SHOES FOR PEOPLE OVER 65

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3,000,000 joggers over 65 × 20% (in Florida) = 600,000
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600,000 × 2% target market share = 12,000 pairs
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You first estimate the total market, then the share you can reasonably capture.

07 Breakeven analysis NOT FOR THE MCQ

NOT FOR THE MCQ

Your course material marks this section (7.7) "**NOT FOR THE MCQ**". It is still useful for the exercises and for understanding how profitability works.

Breakeven analysis: a method of determining the level of sales at which the company will **break even (have no profit or loss)**.

FIXED COSTS
Don't change when the quantity sold changes (rent, salaries, insurance).

VARIABLE COSTS
Vary in total with quantity sold but stay constant per unit (materials, sales commissions).

CONTRIBUTION MARGIN PER UNIT
Selling price per unit – variable cost per unit.

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Breakeven point (units) = fixed costs ÷ (price per unit − variable cost per unit)
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THE BOOK'S EXAMPLE: SHOES AT $80

Fixed costs = 210,000 (salaries) + 60,000 (rent) + 10,000 (advertising) + 8,000 (insurance) + 12,000 (other) = **$300,000**.
Variable cost = $40 (manufacturing) + $5 (commission) = **$45**; contribution margin = 80 − 45 = **$35**.

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Breakeven = 300,000 ÷ 35 = 8,571.4 ≈ 8,571 pairs
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The book rounds to 8,571; strictly, you must sell 8,572 pairs to stop losing money (the calculator rounds up).

With 12,000 pairs expected, the book computes (12,000 − 8,571) × 35 = **$120,015** profit; the exact calculation 12,000 × 35 − 300,000 gives $120,000 (the gap comes from rounding the breakeven point).

The "Breakeven calculator" tab lets you change these figures, and the "Worked exercises" tab covers the book's exercises (dog grooming, burgers).

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08 The product development process

The success of a business depends on its ability to **identify the unmet needs** of consumers and to develop products that meet those needs **at a low cost**.

Project team: individuals from **different functional areas** (design, engineering, manufacturing, accounting, finance, marketing) assigned to work together throughout the product development process.

A risky proposition

On average, a company has **twelve failures for every successful product**; Fortune 1000 firms waste over **$60 billion** a year on R&D. A supermarket can choose from about 100,000 items (including 20,000 new products each year) but can stock only 30,000. Three reasons:

- **Trade-offs**: lighter jogging shoes probably won't wear as well; higher quality makes them more costly.
- **Time pressure**: hundreds of decisions made quickly with imperfect information.
- **Economics**: development takes a lot of time and money, and the product must reach the market at the most opportune time; failing to be first can be very costly.

Product development process: the series of activities by which a product idea is transformed into a **final product**.

1. **Evaluate opportunities and select the best product idea**
   A team reviews ideas: does it fill an unmet need? Will enough people buy it? Do we have the resources and expertise to make it?

2. **Get feedback to refine the product concept**
   The **product concept** describes what the product will look like and how it will work; it is tested with potential buyers.

3. **Make sure the product performs and appeals to consumers**
   How it will be made, which components, in-house or outsourced; appealing, safe, easy to use and maintain.

4. **Design with manufacturing in mind**
   High quality at the lowest possible cost: minimize the number of parts and simplify components.

5. **Build and test prototypes**
   A **prototype** is a physical model of the product, refined until the final version is tested by customers.

6. **Ramp up production and run market tests**
   Ramp-up stage: employees are trained in production processes, products are inspected, samples are given to potential customers.

7. **Launch the product**
   Ongoing production and widespread distribution.

09 Artificial intelligence

Artificial intelligence: "the ability of a digital computer or computer-controlled robot to **perform tasks commonly associated with intelligent beings**" (Encyclopædia Britannica).

Machine learning: a subfield of AI, "the science (and art) of programming computers so they can **learn from data**". Its basic structure is the **neural network**, a scaled-down model of part of a human brain.

How a network "learns": apples and oranges

1. You train the network with hundreds of pictures **labeled** "apple" or "orange": the **training set**.
2. You then show it new pictures: the **test set**.
3. If it identifies them correctly, it has "learned"; otherwise, more training and testing are needed. Telling dogs from cats takes thousands or even millions of pictures.

FOUR TAKEAWAYS

1. A computer does not learn **without a person** who sets up the training and testing.
2. You need to **define your goal** before training.
3. Inaccurate information gives faulty results: **garbage in, garbage out**.
4. Learning can require extensive training and testing, but computers operate at electronic speed.

In a **Forbes** survey of more than 300 top executives, **95%** said AI will play an important role in their responsibilities. Benefits cited: increased **productivity**, reduced operating **costs**, improved **speed to market**, and transformed business and operating models. In medicine, Google's AI detected lung and breast cancer better than radiologists.

CATEGORY 1: AI TO FEAR

AI **automates repetitive, often boring work**: fast food, retail, agriculture, warehousing, then retail sales, trucking, manufacturing and repetitive office work.

CATEGORY 2: AI TO ADMIRE

AI helps employees become **more effective**: jobs requiring higher education, judgment and social skills, protected in the near term.

10 Protecting your idea: the patent

Patent, granted by the **USPTO** (U.S. Patent and Trademark Office): it gives "the **right to exclude others from making, using, offering for sale, or selling**" the invention in the United States for **twenty years**.

Three criteria, all required

NEW

No one else can have known about it, used it or written about it before you filed.

NOT OBVIOUS

Sufficiently different from everything used for that purpose in the past.

HAS UTILITY

It must have some value; it can't be useless.

- **Document** your idea as soon as you have it and have it witnessed.
- The formal application process takes about **two years**.
- A U.S. patent protects only in the United States: you need a patent **in each country**.

FIGURES FROM THE BOOK

PowerSki took out more than **80 patents** in three years. IBM received **8,682** patents in 2021 (29th straight year at the top). The USPTO issued about **374,000** patents in 2021; the number has doubled over the last decade.

11 The Nike case

The athletic footwear industry

The book traces the history of the industry: **Charles Goodyear**'s **vulcanization** of rubber, then the jogging boom driven by **Bill Bowerman** (Nike co-founder). Today, **ath-leisure** (athletic shoes worn every day) accounts for about **80%** of athletic shoes sold.

Innovation at Nike

- The **Innovation Kitchen** and the **Sport Research Lab** study movement and materials.
- **Category product teams** (running, basketball…) bring together designers, engineers and marketers.
- About **300** shoe models a year; about **17 months** from drawing board to store shelf.
- Nike has also had missteps: innovation stays risky even for a leader.

LINK TO THE CHAPTER

You find the development steps again: internal ideas, a cross-functional project team, prototypes and tests, then launch.

Revision sheet based on the textbook Exploring Business (FlatWorld, version 5.0.3), chapters 7 and 8. Sections marked "not for the MCQ" are labeled "NOT FOR THE MCQ" in the course material; worked answers are suggestions.